

A few salient features of Git are:

- Free and open source
- Inexpensive branching and merging
- Distributed in nature
- Implicit backup
- Small, fast and less resource hungry

Basic terminology

This section discusses the terminology and operations of Git.

Working tree: This is a directory on the local file system where the source code resides.

Repository: Repository is the workspace that contains source code and metadata of its versions. The local repository resides on the local file system, whereas the remote repository resides on the remote Git server.

Commit: This is a Git object that represents the change set within the repository. A unique SHA-1 hash is assigned to each commit.

Clone: This is a Git operation that downloads the entire remote repository to the local file system.

Pull: This syncs the local repository with the remote repository by downloading new files.

Push: This Git operation uploads files (one or more commits) from a local repository to the remote repository.

Installation and configuration

Git is free and open source software. One can download and use it without any restrictions. Follow the steps given below to install Git on a Windows server:

1. Go to <https://gitforwindows.org/> and download the latest version of Git.
2. Double click on the installer and follow on-screen instructions.
3. Don't forget to select the 'Windows Explorer Integration' option during setup. Please refer to Figure 1.
4. Go to the *Start* menu and launch 'Git Bash'. You should see a command prompt, as shown in Figure 2.

We have successfully installed Git on a Windows machine. Let us configure it using CLI. Configure the

user name and email by executing the commands given below in a terminal:

```
$ git config --global user.name "Jarvis"
$ git config --global user.email
"jarvis@acme.com"
```

Note: Use your user name and e-mail address while configuring Git. This information will appear in each commit you perform.

Now, execute the command given below to validate the configuration:

```
$ git config -list
```

This command will generate the following output:

```
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
```

```
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/
mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
user.name=Jarvis
user.email=jarvis@acme.com
```

Here, we can see the user name and email address, which we configured earlier. Congrats! You have successfully configured Git on your machine.

Basic Git workflow

Most of the times, you will be using the existing repository hosted in your organisation's network. Hence, we will use a public repository for demonstration. We will walk through

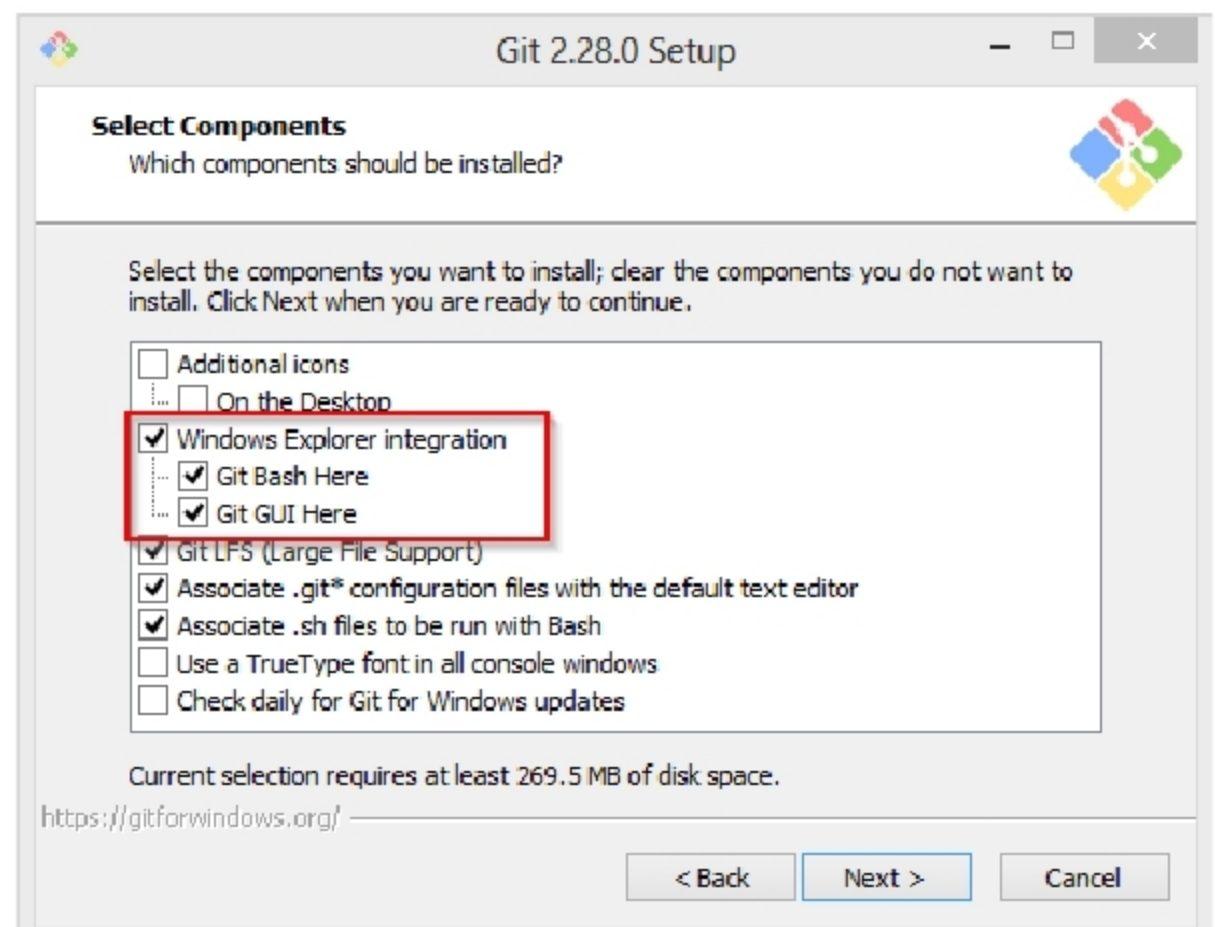


Figure 1: Git setup

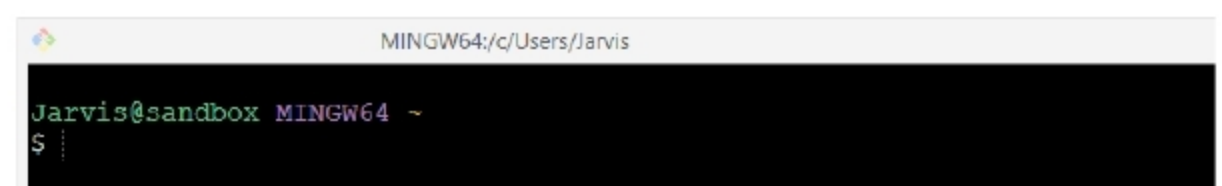


Figure 2: Git Bash